GOVERNEMENT OF KARNATAKA
KARNATAKA SCHOOL EXAMINATION AND ASSEMENT BOARD
6th CROSS, MALLESHWARAM, BENGALURU-560003
2025-26 II PUC MODEL QUESTION PAPER-3

Subject: 34 = Chemistry **Maximum Marks:** 70
Time: 3.00 Hours **No. of Questions:** 46

Instructions

1. Question paper has FIVE parts. All parts are compulsory.
2. a. Part-A carries 20 marks. Each question carries 1 mark.
   b. Part-B carries 06 marks. Each question carries 2 marks.
   c. Part-C carries 15 marks. Each question carries 3 marks.
   d. Part-D carries 20marks. Each question carries 5 marks.
   e. Part-E carries 09 marks. Each question carries 3 marks.
3. In Part-A questions, **first attempted answer** will be considered for awarding marks.
4. Write balanced chemical equations and draw neat labeled diagrams and graphs wherever necessary.
5. Direct answers to the numerical problems without detailed steps and specific unit for final answer will not carry any marks.
6. Use log tables and simple calculator if necessary (use of scientific calculator is not allowed).
7. For a question having circuit diagram/figure/ graph/ diagram, alternate questions are given at the end of question paper in a separate section for visually challenged students.

PART-A

I. Select the correct option from the given choices. **15 × 1 = 15**

1. The common oxidation state shown by the element with atomic number 21 is

   (a) +3 (b) +4 (c) +5 (d) Both +3 and +4

2. Which of the following is homoleptic complex?

   (a) $\left[Co(NH_3)_3Cl_3\right]$ (b) $\left[Co(NH_3)_5Cl\right]Cl_2$

   (c) $\left[Co(NH_3]_4(H_2O)_2\right]Cl_3$ (d) $[Co(NH_3)_6]Cl_3$

3. The correct name of the reaction $R-X + NaI \xrightarrow{\text{Dry acetone}} R-I + NaX$ is

   (a) Finkelstein reaction (b) Wurtz reaction (c) Swarts reaction (d) Friedel –Craft reaction

4. The chemical formula of Hinsberg's reagent is

   (a) C_6H_6SOCl (b) $C_6H_5SOCl_2$ (c) $C_6H_5SO_2Cl$ (d) $C_6H_5SO_3H$

5. Doctor advised a 50-year-old woman enough exposure to sunlight and addition of fish and egg yolk to her diet. Name the vitamin deficiency for the possible disease diagnosed by the doctor.

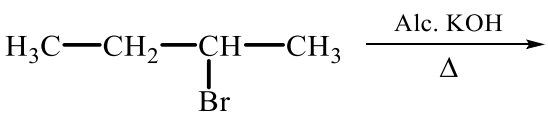
   (a) Vitamin D (b) Vitamin A (c) Vitamin C (d) Vitamin B_{12}

6. Salicylaldehyde can be prepared from which of the following reactants?

   (a) Phenol and Chloroform (b) Phenol, Chloroform and aq. NaOH

   (c) Phenol and CCl_4 (d) Phenol, CCl_4 and NaOH

7. The major product formed in the following reaction is



   (a) 1-butene (b) 2-bntanol (c) 1-bromobutane (d) 2-butene

8. **Statement-I:** In benzylic alcohols, the –OH groups is attached to sp^3 carbon atom next to aromatic ring.
Statement-II: In vinylic alcohols, the –OH group is bonded to a carbon atom of carbon–carbon double bond.
(a) Both Statement I and Statement II are correct.
(b) Statement I is incorrect but Statement II is correct.
(c) Both Statement I and Statement II are incorrect.
(d) Statement I is correct but Statement II is incorrect.

9. For the reaction $2A + 3B + \frac{1}{2}C \longrightarrow P$, the correct rate of the reaction is

(a) $\frac{-d[A]}{dt} = \frac{-2}{3}\frac{d[B]}{dt} = -4\frac{d[C]}{dt}$ (b) $\frac{-d[A]}{dt} = \frac{-3}{2}\frac{d[B]}{dt} = \frac{-1}{4}\frac{d[C]}{dt}$

(c) $\frac{+d[A]}{dt} = \frac{+3}{2}\frac{d[B]}{dt} = \frac{+1}{2}\frac{d[C]}{dt}$ (d) $\frac{+d[A]}{dt} = \frac{+2}{3}\frac{d[B]}{dt} = +4\frac{d[C]}{dt}$

10. Match the terms given in Column I with units given Column II.

Column I	Column II
(i) λ_m	(A) Scm^{-1}
(ii) E_{cell}	(B) cm^{-1}
(iii) κ	(C) Scm^2mol^{-1}
(iv) G^*	(D) V

(a) i-C, ii-D, iii-B, iv-A (b) i-C, ii-D, iii-A, iv-B
(c) i-D, ii-A, iii-C, iv-B (d) i-B, ii-D, iii-A, iv-C

11. Hydration of acetylene (ethyne) with dil. H_2SO_4 and $HgSO_4$ gives
(a) Acetone (b) 2-phenylethanol (c) Acetaldehyde (d) Acetic acid

12. 10 mL of liquid 'A' and 20 mL of liquid 'B' are mixed at 25^oC. The volume of the solution was measured to be 30.1 mL then,
(a) $\Delta H_{mix} > 0$, Solution shows negative deviation from Raoult's low.
(b) $\Delta H_{mix} < 0$, Solution shows negative deviation from Raoult's low.
(c) $\Delta H_{mix} > 0$, Solution shows positive deviation from Raoult's low.
(d) $\Delta H_{mix} < 0$, Solution shows positive deviation from Raoult's low.

13. Arrange the following complexes in the increasing order of crystal field splitting;
$[Co(CN)_6]^{3-}$, $[CoF_6]^{3-}$, $[Co(H_2O)_6]^{3+}$ and $[NiCl_4]^{2-}$
(a) $[NiCl_4]^{2-} < [CoF_6]^{3-} < [Co(H_2O)_6]^{3+} < [Co(CN)_6]$
(b) $[Co(CN)_6]^{3-} < [Co(H_2O)_6]^{3+} < [CoF_6]^{3-} < [NiCl_4]^{2-}$
(c) $[Co(CN)_6]^{3-} < [Co(H_2O)_6]^{3+} < [Nicl_4]^{2-} < [CoF_6]^{3-}$
(d) $[CoF_6]^{3-} < [Nicl_4]^{2-} < [Co(H_2O)_6]^{3+} < [Co(CN)_6]^{3-}$

14. A binary solution has two components 'A' and 'B'. The mole fraction of component 'A' is 0.5, then the number of moles of components 'A and 'B' in the solution is

(a) $n_A > n_B$ (b) $n_A < n_B$ (c) $n_A = n_B$ (d) All of these

15. The reagent which does not react with both, acetone and benzaldehyde.

(a) Sodium hydrogensulphite (b) Phenyl hydrazine (c) Fehling's solution (d) Grignard reagent

II. Fill in the blanks by choosing the appropriate word from those given in the brackets:

(reducing, one, o-cresol, orange, yellow. +3) **5 × 1 = 5**

16. Van't Hoff factor (i) for a non-electrolyte in a solution is _____.
17. The actinoids generally show _____ oxidation state.
18. The colours of p-Hydroxyazobenzene is _____
19. In _____, the –OH group is attached to the carbon that is sp^2 hybridised.
20. The carbohydrates which reduce Fehling's solution and Tollen's reagent are called as _____ sugars.

PART – B

III. Answer any three of the following. Each question carries 2 marks. **3 × 2 = 6**

21. Write any two differences between order and molecularity of a reaction.
22. What are the optically active compounds? Give the condition for the molecule to be optically active.
23. Write the products formed when anisole reacts with ethanoyl chloride in the presence of anhydrous $AlCl_3$ catalyst.
24. With respect to first transition elements series. Which element has

(i) highest second ionisation enthalpy?

(ii) lowest enthalpy of atomisation? **(1+1)**

25. Identify the proteins and give any one difference between these proteins, the protein present in the hair and the protein present in egg albumin.

PART – C

IV. Answer any three of the following. Each question carries 3 marks. **3 × 3 = 9**

26. Write the balanced chemical equations involved in the manufacture of potassium dichromate from chromite ore.
27. Using Valence Bond Theory [VBT]; explain geometry, hybridisation and magnetic property of $[Co(NH_3)_6]^{3+}$ ion. [Given: Atomic mass of cobalt is 27].
28. On treating $CoCl_3.4NH_3$ with excess of $AgNO_3$ solution, 1 mole of AgCl is formed. Formulate the complex and write its geometrical isomers.
29. Write the formula of the three different types of compounds formed when lanthanoids reacts with carbon or heated with carbon?
30. Write the IUPAC name and structure of $\left[Co_2(CO)_8\right]$ and mention the number of bridged CO groups.

V. Answer any two of the following. Each question carries 3 marks. **2 × 3 = 6**

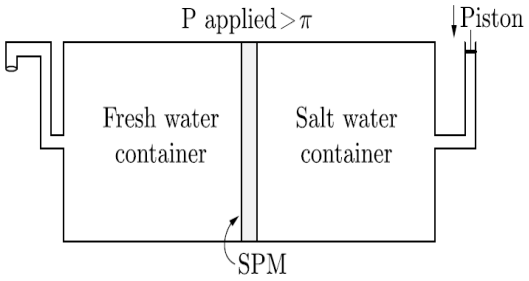
31. Give an example of inert electrode. Write the representation and half- cell reaction for bromine electrode.
32. Write the overall reaction of Ni-Cd battery. Mention the anode and cathode of it.
33. Derive an integrated rate equation for the rate constant of a zero-order reaction.

34. Given below is the sketch of a plant for carrying out a process.

(i) Name the process occurring in the given plant.

(ii) Name any one SPM which can be used in this plant.

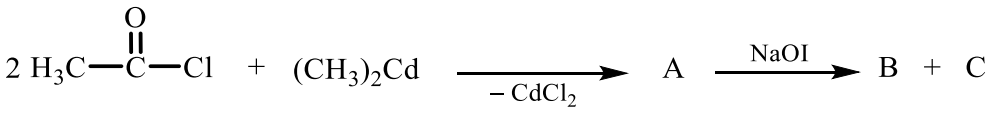
(iii) Give one practical use of the plant.



PART – D

VI. Answer any four of the following. Each question carries 5 marks. **4 × 5 = 20**

35. (a) Identify the compounds 'A', 'B' and 'C' in the following reaction.

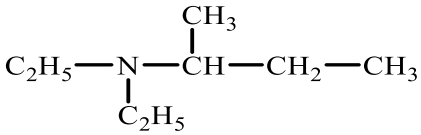


(b) An organic compound 'P' ($C_7H_6O_2$) reacts with thionyl chloride gives product 'Q'. 'Q' on the further undergoes hydrogenation in the presence of Pd-$BaSO_4$ forms compound 'R'. Write the equations involved in the chemical reactions. **(3+2)**

36. (a) Explain Hoffmann bromamide degradation reaction with chemical equation.

(b) Discuss the coupling reaction of Benzene diazonium chloride with aniline. Give equation.

(c) Write the IUPAC name of



(2+2+1)

37. (a) Glucose is a monosaccharide and an aldohexose. Which oxidising agent should be used to bring about oxidation of only the aldehydic group present in glucose? Write the chemical equation.

(b) Mention any two forces which stabilises the 2^0 and 3^0 structures of proteins.

(c) Name the type linkage present in Dinucleotide obtained by joining two nucleotides. **(2+2+1)**

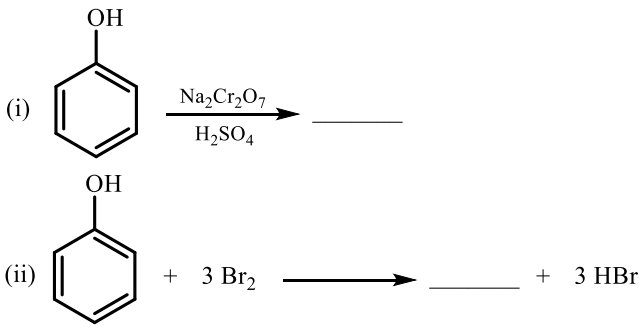
38. (a) Explain Fittig reaction with general equation.

(b) Write the mechanism for the conversion of methyl chloride to methanol.

(c) How many stereo centres are present in Butan-2-ol? **(2+2+1)**

39. (a) Write the steps involved in the mechanism of acid catalysed dehydration of ethanol to ethene.

(b) Complete the following reactions;



(3+2)

40. (a) Explain Hell-Volhard –Zelinsky reaction with chemical equation.

(b) Write the reaction for the conversion of toluene to benzaldehyde using chromyl chloride. Name the reaction. **(2+2+1)**

(c) Name the gas used to facilitate the nucleophilic addition of ethylene glycol to aldehydes and ketones.

PART – E (PROBLEMS)

VII. Answer any three of the following. Each question carries 3 marks. 3 × 3 = 9

41. Calculate the osmotic pressure of 0.5%(w/v) aqueous solution sucrose at 300K. (R = 0.0083 L bar mol^{-1} and molar mass of sucrose = 342 g mol^{-1}).

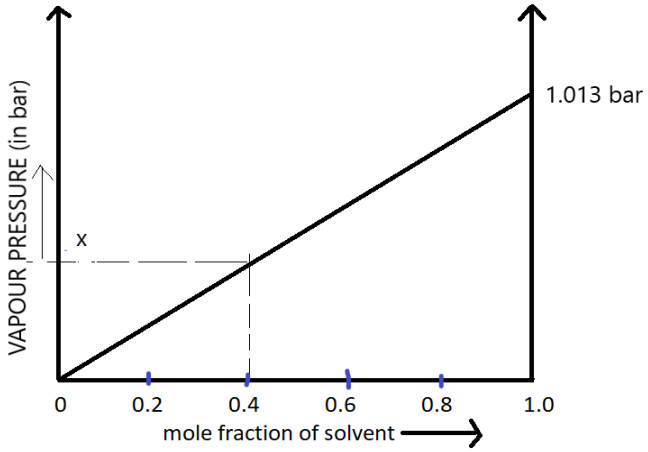
42. Calculate the electromotive force (emf) of the cell. Assume the cell reaction at 25^0 C with the reaction: $Cu_{(s)} + 2Ag^{+}_{(1\times10^{-3}M)} \longrightarrow Cu^{2+}_{(0.250M)} + 2Ag_{(s)}$. $E^0_{cell} = 2.97$ V

43. A column containing 0.05 M NaOH has an area of cross section 0.785 cm^2 and length of 1m shows a resistance of 1.11×10^4 Ω. Calculate the molar conductivity of the solution.

44. A first order reaction has rate constant of 2.3×10^{-5} s^{-1} at 500 K. At what temperature rate constant of the reaction becomes 17.25×10^{-4} s^{-1}? (E_a = 191.472 KJ mol^{-1} and R =8.314 J $K^{-1}mol^{-1}$)

45. A first order reaction has half-life of 23 min. How much time will be needed for 30% completion of the reaction?

46. If a solution obeys Raoult's law for all concentrations, its vapour pressure would vary linearly from zero to the vapour pressure of pure solvent as shown in graph. Calculate the value of 'x'.



PART - F

(For visually challenged students only)

34. Name any three parameters which affects solubility of gas in liquid.

46. The vapour pressure of compound 'A' and compound 'B' at 25^0 C are 90 mm Hg and 15 mm Hg respectively. If 'A' and 'B' are mixed each other such that mole fraction of 'A' in the solution is 0.6. Calculate the mole fraction of 'B' in vapour phase.

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